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| Semester: III | Year : AY 2026-27 |
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| Subject: Programming with Java |
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| ASSIGNMENT No : 5 |
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| TITLE : Collection Classes and StringBuffer |
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### Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

### Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

### Programme

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
TodoList.java
1- import java.util.ArrayList;
2
3- class TodoList {
4-     public static void main(String[] args) {
5         ArrayList<String> tasks = new ArrayList<>();
6
7         tasks.add("Complete Java Assignment");
8         tasks.add("Buy Groceries");
9         tasks.add("Prepare Presentation");
10
11         StringBuffer taskDisplay = new StringBuffer();
12         taskDisplay.append("--To Do List-- \n");
13
14         for (int i = 0; i < tasks.size(); i++) {
15             taskDisplay.append((i + 1)).append(" ").append(tasks.get(i)).append("\n");
16         }
17
18         System.out.println(taskDisplay);
19     }
20 }
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
CourseRegistration.java
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 class CourseRegistration {
5     public static void main(String[] args) {
6         ArrayList<String> courses = new ArrayList<>();
7         Scanner scanner = new Scanner(System.in);
8         boolean running = true;
9
10        while (running) {
11            System.out.println("\n--- Course Registration Menu ---");
12            System.out.println("1. Add Course");
13            System.out.println("2. Remove Course");
14            System.out.println("3. View Courses");
15            System.out.println("4. Exit");
16            System.out.print("Enter choice: ");
17
18            int choice = Integer.parseInt(scanner.nextLine());
19
20            switch (choice) {
21                case 1:
22                    System.out.print("Enter course name to add: ");
23                    String courseToAdd = scanner.nextLine();
24                    courses.add(courseToAdd);
25                    System.out.println("Course added successfully.");
26                    break;
27
28                case 2:
29                    System.out.print("Enter course name to remove: ");
30                    String courseToRemove = scanner.nextLine();
31                    if (courses.remove(courseToRemove)) {
32                        System.out.println("Course removed successfully.");
```

```

30         String courseToRemove = scanner.nextLine();
31         if (courses.remove(courseToRemove)) {
32             System.out.println("Course removed successfully.");
33         } else {
34             System.out.println("Course not found.");
35         }
36         break;
37
38         case 3:
39             StringBuffer sb = new StringBuffer();
40             sb.append("\n--- Registered Courses ---\n");
41             for (int i = 0; i < courses.size(); i++) {
42                 sb.append((i + 1)).append(". ").append(courses.get(i)).append("\n");
43             }
44         }
45         System.out.println(sb);
46         break;
47
48         case 4:
49             running = false;
50             System.out.println("Exiting system.");
51             break;
52
53         default:
54             System.out.println("Invalid choice. Try again.");
55             break;
56     }
57 }
58 scanner.close();
59 }
60 }

```

## Output

### Program 1 output:

| Output                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------|
| <pre> --To Do List-- 1. Complete Java Assignment 2. Buy Groceries 3. Prepare Presentation  === Code Execution Successful === </pre> |

## Program 2 output:

| Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>--- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 1 Enter course name to add: DS Course added successfully.  --- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 1 Enter course name to add: OS Course added successfully.  --- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 3  --- Registered Courses --- 1. DS 2. OS</pre> | <pre>--- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 2 Enter course name to remove: OS Course removed successfully.  --- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 3  --- Registered Courses --- 1. DS  --- Course Registration Menu --- 1. Add Course 2. Remove Course 3. View Courses 4. Exit Enter choice: 4 Exiting system.  === Code Execution Successful ===</pre> |

## Conclusion

These programs demonstrate the combined usage of Java's ArrayList and StringBuffer classes. The ArrayList provides dynamic array capabilities to easily store, modify, and manage elements like tasks or registered courses. Meanwhile, StringBuffer efficiently builds formatted output text without creating extra string objects in memory. The Course Registration System uses a switch statement inside a loop to provide an interactive menu for user operations without extra class methods.